LESSON 01 / 07

## Counting is matching, construction is multiplication

To count a set $X$ means to find $|X|$. Two master moves solve almost everything: **(A) construct** each object by a sequence of choices (multiply), **(B) match up** $X$ bijectively with a set already counted (inherit its size). Every formula in this chapter is one of these, packaged.

KEY RESULT

### Product rule and sum rule (with their exact hypotheses)

**Product rule:** a task in successive steps, step $i$ offering $n_i$ choices *regardless of earlier choices*, has $n_1 n_2 \cdots n_k$ outcomes. The independence clause is the hypothesis: "5 choices then 4 choices" for choosing 2 people from 5 counts **ordered** pairs (20), because $\{A, B\}$ is built twice.

**Sum rule:** if $X$ is split into disjoint, exhaustive cases, $|X|$ is the sum of the case counts. Cases must be *every case counted once, every object in some case*.

KEY RESULT

### Complement: negatives are rigid, positives are flexible

"At least one" / "no" / "not all" ⇒ count the complement and subtract from the total. Committees of 5 from 12 men and 8 women with at least one woman: $\binom{20}{5} - \binom{12}{5} = 15504 - 792 = 14712$, instead of five direct cases. Strings of length 4 over $\{0,1,2,3,4\}$ containing a 0: $5^4 - 4^4 = 625 - 256 = 369$.

WORKED EXAMPLE

### Worked — cases where steps interact

4-digit numbers with distinct digits divisible by 5: split on the units digit (the thousands digit's options depend on it). Units 0: $9 \cdot 8 \cdot 7 = 504$. Units 5: $8 \cdot 8 \cdot 7 = 448$. Total 952. Splitting is the cure when independence fails in exactly one place.

INTUITION

### Bijections you already own

Subsets of an $n$-set ↔ binary strings of length $n$ (in/out per element): $2^n$ subsets. Increasing functions $f$ with $f(1) < f(2) < f(3)$ into $\{1, \ldots, 5\}$ ↔ 3-subsets: $\binom{5}{3} = 10$. Pairs $A \subseteq B$ of subsets: each element has 3 states (in neither, in $B$ only, in both): $3^n$. Monotone/ordered-with-condition almost always means: find the unordered object it encodes.

COMMON TRAP

### The three sins of counting (check every answer against them)

1. **Overcounting:** one object built by two choice paths. 2. **Undercounting:** cases not exhaustive. 3. **Broken independence:** multiplying counts that are not fixed per step. Cure for all three: verify a small case by listing. Also ask: objects labelled or unlabelled? order matters or not? is the answer a plausible size (never more than the total pool)?

ASSUMPTIONS & CONVENTIONS

### Standing conventions

$n, r$ are non-negative integers with $0 \le r \le n$ unless stated; $\binom{n}{r} = 0$ if $r < 0$ or $r > n$ (a useful convention in sums); $0! = 1$ (exactly one empty arrangement). "Arrangements" are ordered; "selections/committees" are unordered.

LESSON 02 / 07

## Ordered selections

SHORT DERIVATION

**nPr from filling labelled roles**

Fill $r$ labelled positions from $n$ distinct objects: $n$ options, then $n-1$, ..., then $n-r+1$ (each count fixed regardless of history). Hence

$$P(n,r)=\frac{n!}{(n-r)!},\qquad P(n,n)=n!$$

Equivalently $P(n,r)$ counts injective functions from an $r$-set to an $n$-set; $n!$ counts bijections of an $n$-set. $0!=1$ is forced: there is exactly one empty arrangement.

KEY RESULT

**Identical objects: label, count, divide by the symmetry**

Arrangements of a multiset with $n_1,\dots,n_k$ copies of $k$ types ($n=\sum n_i$):

$$\frac{n!}{n_1!\,n_2!\cdots n_k!}$$

Proof: temporarily label all copies ($n!$ words); each unlabelled word is hit exactly $n_1!\cdots n_k!$ times (permute labels within each type). Division is legal **iff every fibre has the same size** — the exact-symmetry condition; where it fails (necklaces with identical beads), Part II's Burnside lemma is the fix.

Words from COMBINATORICS (C, O, I twice each, 13 letters): $\frac{13!}{2!\,2!\,2!}=778377600$.
MATHEMATICS: $\frac{11!}{2!\,2!\,2!}=4989600$.

WORKED EXAMPLE

**Worked — bundles inside identical-letter words**

Words from DAAR with the two A's adjacent: bundle AA into one super-object; arrange $\{D, AA, R\}$: $3!=6$ (the identical A's give the bundle one internal order). Words: DAAR, AADR, ADAR, ADRA, ARDA, RADA ✓.

COMMON TRAP

**Common traps — permutations**

- Dividing by symmetry when the symmetry is not exact (some arrangements fixed by the swap).
- "Choose first, then second" silently counting ordered pairs for an unordered question — off by $r!$.
- Bundles: internal orders multiply only if the bundled objects are distinct; identical objects contribute 1.
- $P(n,r)$ with $r>n$ is 0, not a formula to evaluate.

LESSON 03 / 07

## Around a table

KEY RESULT

**Circular permutations: $(n-1)!$**

$n$ distinct people around a round table, seatings identified up to rotation: $(n-1)!$. Two proofs: anchor one person as reference and arrange the rest linearly; or divide $n!$ by the $n$ rotations (every orbit has size exactly $n$ for distinct people). If flips are also identified (necklaces of **distinct** beads, $n\geq 3$): $\frac{(n-1)!}{2}$.

Two specified people adjacent (round table): bundle, $2(n-2)!$; not adjacent: $(n-1)!-2(n-2)!=(n-3)(n-2)!$. For 10 people with A, B non-adjacent: $9!-2\cdot 8!=362880-80640=282240$.

## The three restriction methods (linear arrangements)

KEY RESULT

**Bundle, gap, relative order — pick by the wording**

- **"Together/consecutive":** bundle the $k$ objects into one super-object ($k!$ internal orders if distinct), arrange the $n-k+1$ objects.
- **"No two adjacent":** gap method. Arrange the $m$ ordinary objects ($m!$), creating $m+1$ gaps; place the $k$ special objects into **distinct** gaps: $\binom{m+1}{k}$ choices and $k!$ orders: total $m!\binom{m+1}{k}k!$. Six people with $A$, $B$, $C$ pairwise non-adjacent: $3!\cdot\binom{4}{3}\cdot 3! = 144$.
- **"Fixed relative order" (A before B before C, not necessarily adjacent):** among all $n!$ orders the $k$ specified objects show each of their $k!$ relative orders equally often: answer $\frac{n!}{k!}$. "$A$ before $B$" alone: $\frac{n!}{2}$.

WORKED EXAMPLE

**Worked — composing methods**

Six people $P$, $Q$, $R$, $S$, $T$, $U$ in a row; $P$, $Q$ together and $R$ not at an end. Bundle $PQ$: $2\cdot 5! = 240$ arrangements of the $5$ objects. Subtract $R$ at an end: $2$ ends, then $4!$ arrangements of the rest, bundle order $2$: $2\cdot 4!\cdot 2 = 96$. Answer $240-96 = 144$.

Order of operations: bundle first, then apply the positional restriction to the shrunken list.

Alternation: $5$ men and $5$ women alternating in a row: $2$ patterns $\times 5!\cdot 5! = 28800$.

COMMON TRAP

**Common traps — restrictions**

- Gap method requires the ordinary objects arranged **first**, and needs $k \le m+1$ to be possible at all.
- "Not adjacent" for a specified pair is bundle-and-subtract; "no two of a set adjacent" is the gap method. Swapping them is the classic error.
- Circular "not adjacent" via linear gap method: wrong geometry — use total minus bundled.
- Necklace division by 2 needs distinct beads (or Burnside); identical beads break exactness.

LESSON 04 / 07

## Unordered selections

SHORT DERIVATION

**Why dividing by r! is exact here**

Every $r$-subset generates exactly $r!$ orderings, and every ordering comes from one subset: the ordered selections $P(n,r)$ split into equal fibres of size $r!$. So

$$\binom{n}{r} = \frac{n!}{r!\,(n-r)!}, \qquad \binom{n}{r} = \binom{n}{n-r}$$

The symmetry is complementation: choosing what to keep = choosing what to discard. First question of every problem: do the objects come with positions/labels (permutations) or not (combinations)? Mixed problems do both, in that order.

KEY RESULT

**Pascal and hockey-stick (proofs are one split each)**

**Pascal:** $\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}$ — split $r$-subsets by whether a fixed element is in (then choose $r-1$ more) or out (choose all $r$ from the rest).

**Hockey-stick:** $\sum_{i=r}^{n} \binom{i}{r} = \binom{n+1}{r+1}$ — count $(r+1)$-subsets of $\{1, \ldots, n+1\}$ by their largest element $i+1$: the rest is an $r$-subset of $\{1, \ldots, i\}$. E.g. $\binom{1}{1} + \binom{2}{1} + \binom{3}{1} + \binom{4}{1} = 10 = \binom{5}{2}$ (which is also $1+2+3+4$).

WORKED EXAMPLE

**Worked — restricted selections**

At least one of 4 special books among 5 chosen from 20: $\binom{20}{5} - \binom{16}{5} = 15504 - 4368 = 11136$ (complement again).

Team of 11 from 7 batsmen, 5 bowlers, 3 all-rounders, at least 3 batsmen and at least 2 bowlers: enumerate by all-rounders $a = 0..3$ and sum products $\binom{7}{b}\binom{5}{w}\binom{3}{a}$ over valid $(b, w)$: $12 + 198 + 660 + 490 = 1360$. The template: one free variable plus bounds makes a finite table — check boundary triples twice, and **do not drop the $\binom{3}{a}$ factor** for the enumerated group (the classic table error: rows for $a = 1, 2$ are each 3 times their unweighted subsums).

| | |
|---|---|
| n=0 | 1 |
| n=1 | 1 , 1 |
| n=2 | 1 , 2 , 1 |
| n=3 | 1 , 3 , 3 , 1 |
| n=4 | 1 , 4 , 6 , 4 , 1 |
| n=5 | 1 , 5 , 10 , 10 , 5 , 1 |

Pascal's triangle — row n of (a+b)ⁿ, each entry the sum of the two above

COMMON TRAP

**Common traps — combinations**

- $\binom{n}{r}$ computed for "arrange the chosen few" problems (or vice versa): decide order first.
- Case splits on "at least" that double-count overlaps — use complement or inclusion–exclusion (sheet 06).
- Boundary triples dropped in group-selection tables (the $(7, 3)$-type corner cases).
- Choosing from groups: multiply the independent group choices, then sum over the valid compositions — not the reverse.

LESSON 05 / 07

## Identical objects into distinct boxes

SHORT DERIVATION

**The stars-and-bars bijection**

Distributing $n$ **identical** balls into $k$ **distinct** boxes is the same data as a count tuple $(x_1, \ldots, x_k)$, $x_i \geq 0$, $\sum x_i = n$ — and the same as a word of $n$ stars and $k-1$ bars (stars for box 1, bar, stars for box 2, ...). Choosing the bar positions among $n+k-1$ symbols:

$$\binom{n+k-1}{k-1} \quad \text{(empty boxes allowed)}$$

$$\binom{n-1}{k-1} \quad (\text{every box} \geq 1)$$

The positive case: pre-place one ball in each box, distribute the remaining $n-k$ freely. Non-negative solutions of $x+y+z=15$: $\binom{17}{2} = 136$; positive: $\binom{14}{2} = 91$ (difference $45 = 42 + 3$: exactly one zero $3 \cdot 14$, exactly two zeros $3$ ✓).

KEY RESULT

**The same formula in four costumes**

- **Multisets:** choosing 12 fruits from 5 varieties (counts only): $\binom{12+5-1}{4} = \binom{16}{4} = 1820$.
- **Compositions:** ordered sums of $n$ into $k$ positive parts: $\binom{n-1}{k-1}$ (separators in the $n-1$ gaps between unit blocks); any number of parts: $2^{n-1}$. Ordered sums of 10 into parts $\geq 2$: shift $b_i = a_i - 1$, sum over the number of parts $k = 1..5$: $1 + 7 + 15 + 10 + 1 = 34$ (the $k = 1$ term is the single-part sum $[10]$ itself — easy to drop).
- **Non-negative solutions with lower bounds:** shift each variable down to 0 first.
- **Upper bounds:** complement/inclusion–exclusion: $x_1 + x_2 + x_3 + x_4 = 20$, $x_1 \leq 5$: total $\binom{23}{3} = 1771$ minus ($x_1 \geq 6$, shift) $\binom{17}{3} = 680$ gives 1091.

COMMON TRAP

**Common traps — stars and bars**

- Identical vs distinct balls: distinct balls into boxes is $k^n$ (functions), a completely different count. Read which objects are identical.
- Applying $\binom{n+k-1}{k-1}$ to ordered compositions with positive parts (that is $\binom{n-1}{k-1}$).
- Upper-bound problems with two violated bounds need genuine inclusion–exclusion, not one subtraction.
- Balls identical + boxes identical (partitions) has **no closed form** — Part II; do not invent one.

LESSON 06 / 07

## The principle

SHORT DERIVATION

**Inclusion–exclusion, proved by tracking one element**

For bad events $A_1, \ldots, A_n$ in a universe $X$:

$$\left|X \setminus \bigcup A_i\right| = \sum_{S \subseteq \{1..n\}} (-1)^{|S|} \left|\bigcap_{i \in S} A_i\right|$$

Proof: an element lying in exactly $t \geq 1$ of the sets is counted $\sum_{j=0}^{t} (-1)^j \binom{t}{j} = (1-1)^t = 0$ times; an element in none is counted once. **Universal recipe:** define bad events; compute the intersection size for a generic set of $k$ specified events (usually $\binom{n}{k} \times$ something free); write the alternating sum.

Integers 1–1000 divisible by 2, 3 or 5: $500 + 333 + 200 - 166 - 100 - 66 + 33 =$ (so divisible by none: 266).

KEY RESULT

**Derangements: the flagship application**

$D_n$ = permutations of $\{1..n\}$ with **no** fixed point. With $A_i$ = "$i$ is fixed", a specified set of $k$ fixed points leaves $(n-k)!$ free permutations:

$$D_n = \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)! = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}$$

$D_0 = 1, D_1 = 0, D_2 = 1, D_3 = 2, D_4 = 9, D_5 =$ . Also $D_n = (n-1)(D_{n-1} + D_{n-2})$ (where does element 1 go? then whether its partner swaps back), and $D_n$ is the **nearest integer to** $n!/e$ since the tail beyond $n$ is $< \frac{1}{n+1} < \frac{1}{2}$.

Exactly $k$ fixed points: $\binom{n}{k} D_{n-k}$ (for $n = 5$, exactly two correct letters: $\binom{5}{2} \cdot 2 = 20$; check: $44 + 20 + 45 + 10 + 0 + 1 = 120 = 5!$ ✓).

WORKED EXAMPLE

**Worked — onto functions (IE in disguise)**

Onto functions from a 5-set to a 3-set: total $3^5$ minus those missing a value: $3^5 - \binom{3}{1}2^5 + \binom{3}{2}1^5 = 243 - 96 + 3 = 150$.

Hands of 6 cards containing all four suits: $\binom{52}{6} - 4\binom{39}{6} + 6\binom{26}{6} - 4\binom{13}{6} = 8682544$.

COMMON TRAP

**Common traps — inclusion–exclusion**

- Sign alternation starting wrong: the empty intersection (whole universe) carries +.
- Intersection sizes computed for *exactly* $k$ events instead of *at least these* $k$ (IE wants the latter).
- Bounded stars-and-bars terms with negative top: $\binom{m}{k-1}$ for a negative-sum equation is **0** — terminate the sum, never evaluate.
- Derangement problems with only *some* positions forbidden: restrict the event set (then it is partial derangement / rook theory, Part II flavour).

LESSON 07 / 07

## Existence by averaging

KEY RESULT

**Pigeonhole principle (the art is the boxes)**

$n+1$ objects in $n$ boxes ⇒ some box has $\geq 2$. The theorem is trivial; every application is the choice of the box map so that "same box" gives the conclusion.

**Odd-part box:** from any $n+1$ integers in $\{1, \ldots, 2n\}$, two share the same odd part in $x = 2^a \cdot m$ ($m$ odd) — only $n$ odd parts exist — so one divides the other. Sharp: $\{n+1, \ldots, 2n\}$ has no such pair.

**Geometry box:** 5 points in a unit square: quarter it into 4 subsquares; two points share one, at distance $\leq \frac{\sqrt{2}}{2}$ (its diagonal).

## Counting by last-step decomposition

KEY RESULT

**Recurrences: split by the last block**

If objects of size $n$ end with one of a few block types, leaving a smaller instance of the same problem, the counts satisfy a recurrence. $2 \times n$ domino tilings: ends vertical ($a_{n-1}$) or two horizontal ($a_{n-2}$): $a_n = a_{n-1} + a_{n-2}$, $a_0 = a_1 = 1$, so $a_n = F_{n+1}$ (tilings of $2 \times 10$ : 89). Binary strings with no consecutive 1s and subsets with no two consecutive give the same recurrence ($F_{n+2}$). Staircase with steps 1–2: $a_{10} = 89$; steps 1–3: tribonacci. Guard the base cases — a correct recurrence with wrong seeds is a wrong answer.

KEY RESULT

**Recap — the chapter on one card**

Product/sum/complement; bijection = same count. $P(n,r) = \frac{n!}{(n-r)!}$; identical objects $\frac{n!}{\prod n_i!}$; circular $(n-1)!$ (necklace $\frac{(n-1)!}{2}$ distinct beads). Bundle / gap $m!\binom{m+1}{k}k!$ / relative order $\frac{n!}{k!}$. $\binom{n}{r} = \frac{n!}{r!\,(n-r)!}$; Pascal; hockey-stick. Stars and bars $\binom{n+k-1}{k-1}$ / positive $\binom{n-1}{k-1}$; compositions $2^{n-1}$. IE alternating sum; $D_n = n!\sum\frac{(-1)^k}{k!} \approx n!/e$; onto $= \sum(-1)^j\binom{k}{j}(k-j)^n$. Pigeonhole: design the boxes. Recurrence: split by last block, mind the seeds.

COMMON TRAP

**Pre-submission checklist**

Order matters? Labels? Identical objects — is my division by symmetry exact (equal fibres)? Cases disjoint and exhaustive? Did "at least" get complemented? Bounds handled by shift/IE, with negative-top terms set to 0? Small case listed by hand? Answer size sane against the total pool?

INTUITION

**Forward map**

Binomial identities are this chapter counted twice (chapter 9); lattice paths and Catalan, Burnside–Pólya, partitions and Stirling numbers, intersecting families: Part II here. The through-line never changes: construct, split, or match up.

LESSON 01 / 03

ASSUMPTIONS & CONVENTIONS

**STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)**

Lattice-path methods, Catalan numbers and reflection arguments are beyond the official P&C syllabus lines; they are the standard Olympiad continuation of $\binom{n}{r}$ as a path count.

KEY RESULT

**Paths are binomial coefficients**

Monotone paths from $(0,0)$ to $(a,b)$ (steps right/up): choose which steps are up: $\binom{a+b}{b}$. Paths that never rise above the diagonal $y=x$ to $(n,n)$: the **Catalan number**

$$C_n = \binom{2n}{n} - \binom{2n}{n+1} = \frac{1}{n+1}\binom{2n}{n} = 1, 1, 2, 5, 14, 42, \ldots$$

**Reflection principle:** a "bad" path first touching $y=x+1$, reflected up to that point, bijects with all paths from $(-1,1)$ to $(n,n)$ — $\binom{2n}{n-1}$ of them. Subtract. Paths to $(4,4)$ staying weakly below the diagonal: $C_4 = 70 - 56 = 14$.

KEY RESULT

**Ballot theorem**

Votes $p$ for A, $q$ for B, $p>q$: counting orders where A is **strictly ahead throughout**: $\frac{p-q}{p+q}\binom{p+q}{q}$. For $p=5$, $q=3$: $\frac{2}{8}\cdot 56 = 14$.

The same reflection gives paths to $(a,b)$, $a \geq b$, never above the diagonal: $\frac{a-b+1}{a+1}\binom{a+b}{b}$.

INTUITION

**Why Catalan counts everything**

Dyck paths, balanced parentheses, full binary trees, polygon triangulations, non-crossing matchings share one decomposition: *first return splits the object into inside × outside*, giving $C_{n+1} = \sum_{i=0}^{n} C_i C_{n-i}$. Recognising the convolution shape is the solution; the closed form follows. Brute-force checks: Dyck paths of semilength $n \leq 7$ counted by enumeration in math_checks.

COMMON TRAP

**Common traps — path counts**

- "Never above" vs "never below" and weak vs strict inequalities change which line to reflect in ($y=x+1$ vs $y=x-1$): draw the first forbidden point.
- Ballot needs $p>q$ strictly; ties throughout (weakly ahead) is the Catalan-style variant with a different formula.

LESSON 02 / 03

ASSUMPTIONS & CONVENTIONS

**STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)**

Group-action counting is university/Olympiad material. It exists in this chapter because Part I's "divide by the symmetry" is exactly the special case where every orbit has the same size — and fails precisely when it does not.

SHORT DERIVATION

**Burnside's lemma (double counting in disguise)**

A group $G$ acts on colourings $X$; orbits = essentially-different colourings. Count pairs $(g, x)$ with $gx = x$ two ways: by $g$ it is $\sum_g |\mathrm{Fix}(g)|$; by $x$ it is $\sum_x |\mathrm{Stab}(x)| = \sum_x \frac{|G|}{|\mathrm{orbit}(x)|}$, which sums to $|G| \times$ (number of orbits). Hence

$$\#\text{orbits} = \frac{1}{|G|} \sum_{g \in G} |\mathrm{Fix}(g)|$$

A colouring is fixed by $g$ iff it is constant on $g$'s cycles: $k^{\#\text{cycles}}$ colourings in $k$ colours.

WORKED EXAMPLE

**Worked — necklaces and bracelets**

Necklaces of $n$ beads, $k$ colours (rotations only): $N(n,k) = \frac{1}{n} \sum_{d \mid n} \varphi(d)\, k^{n/d}$ (rotations grouped by $\gcd(n,s)$).
$N(4,2) = \frac{1}{4}(16 + 4 + 4) = 6$: RRRR, RRRB, RRBB, RBRB, RBBB, BBBB ✓. Bracelets of 6 beads, 2 colours (full dihedral group of order 12): fixed counts $64 + 2 \cdot 2 + 2 \cdot 4 + 8 + 3 \cdot 16 + 3 \cdot 8 = 156$, orbits $156/12 = 13$ ✓.

COMMON TRAP

**Common traps — Burnside**

- Forgetting the identity element (usually the largest Fix count).
- Grouping rotations wrongly: rotation by $s$ on $n$ beads has $\gcd(n,s)$ cycles, and there are $\varphi(n/d)$ rotations with a given divisor structure — recount small cases by listing (as with $N(4,2) = 6$).
- Applying "divide by $|G|$" directly to colourings: that is exactly the non-exact symmetry error Burnside repairs.

LESSON 03 / 03

ASSUMPTIONS & CONVENTIONS

**STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)**

Integer partitions, Stirling/Bell numbers and extremal combinatorics (Sperner, Erdős–Ko–Rado) are Olympiad territory beyond the JEE P&C lines.

KEY RESULT

**Integer partitions: no closed form, one generating function**

$p(n)$ counts partitions of $n$ (order ignored): $1, 2, 3, 5, 7, 11, 15, 22, 30, 42$ for $n = 1..10$. There is no stars-and-bars model; the count lives in

$$\sum_{n \geq 0} p(n) x^n = \prod_{k \geq 1} \frac{1}{1 - x^k}$$

**Euler:** partitions into **distinct** parts = partitions into **odd** parts, since $\prod (1 + x^k) = \prod \frac{1 - x^{2k}}{1 - x^k} = \prod_{m \text{ odd}} \frac{1}{1 - x^m}$ (for $n = 10$: 10 of each, listed in math_checks by enumeration). Conjugation of Young diagrams: partitions into at most $k$ parts ↔ partitions with largest part $\leq k$. Euler's pentagonal theorem gives $p(n) = p(n-1) + p(n-2) - p(n-5) - p(n$ (check $p(10) = 30 + 22 - 7 - 3 = 42$ ✓).

KEY RESULT

**Stirling numbers of the second kind and Bell numbers**

$S(n,k)$ = partitions of an $n$-set into $k$ nonempty **unlabelled** blocks: recurrence ("where does element $n$ go?") $S(n,k) = kS(n-1,k) + S(n-1,k-1)$; closed form by IE $S(n,k) = \frac{1}{k!}\sum_j (-1)^j \binom{k}{j}(k-j)^n$, because $k!\,S(n,k)$ counts onto functions. $S(6,3) = 3\cdot 25 + 15 = 90 = \frac{729-192+3}{6}$ ✓. Bell numbers $B_n = \sum_k S(n,k)$: $1, 1, 2, 5, 15, 52, 203$. Dividing 6 people into 3 unlabelled pairs: $\frac{6!}{(2!)^3 3!} = 15$.

KEY RESULT

**Extremal families: Sperner and Erdős–Ko–Rado**

**Sperner:** an antichain of subsets of an $n$-set (no one containing another) has at most $\binom{n}{\lfloor n/2 \rfloor}$ members — the middle layer is extremal (LYM proof: a random maximal chain meets an antichain at most once, and hits a $k$-set with probability $1/\binom{n}{k}$).

**Erdős–Ko–Rado:** for $n \geq 2k$, an intersecting family of $k$-subsets of an $n$-set has at most $\binom{n-1}{k-1}$ members; the star (all sets through one fixed point) attains it. Hypothesis essential: for $n < 2k$ every two $k$-sets meet and the whole $\binom{n}{k}$ is intersecting.

COMMON TRAP

**Common traps — advanced counting**

- Confusing partitions (no formula) with compositions ($\binom{n-1}{k-1}$): "order ignored" changes the universe, not just the count.
- $S(n,k)$ vs onto functions: the factor $k!$ is the labelling of blocks.
- Quoting EKR/Sperner without their hypotheses ($n \geq 2k$; antichain, not intersecting).